Yaakkai Leaf Annotator: Annotator Guide

12 October 2026

1. Before you start

This guide shows you how to annotate palm leaf images in the Yaakkai Leaf Annotator, step by step.

You need:

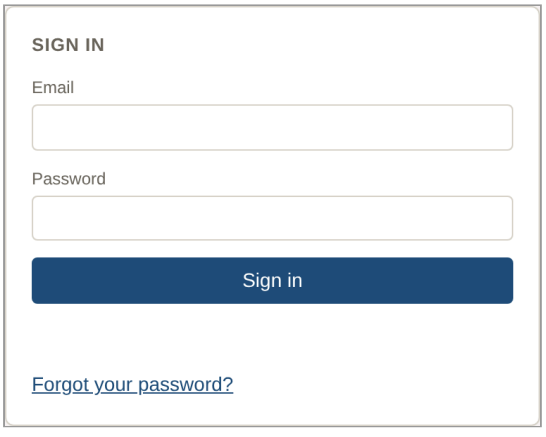
- a laptop or desktop computer (a phone screen is too small for this work)
- Chrome, Edge or Firefox
- the website address and your email and temporary password, given to you by the admin

Your work saves automatically while you work. You never need to press Save.

The screenshots in this guide use a low-resolution sample leaf, so the image looks blurred. Your real leaf photographs will be much sharper. The boxes drawn on the sample are only examples of how the tool works.

2. Signing in and your account

Open the website address. Type your email and password and press **Sign in**.



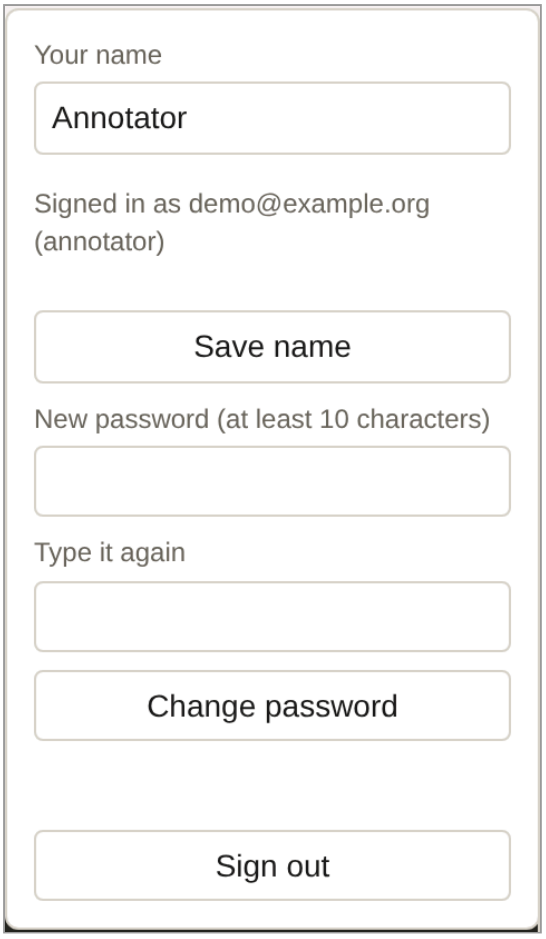
The sign-in screen

The first time you sign in, change the temporary password straight away:

1. Click your name at the top right of the screen. This opens the **Account** menu.

2. Type a new password of at least 10 characters in **New password**, and again in **Type it again**.
3. Press **Change password**.

In the same menu you can change the name other people see next to your work (**Your name**, then **Save name**), and **Sign out** when you finish.

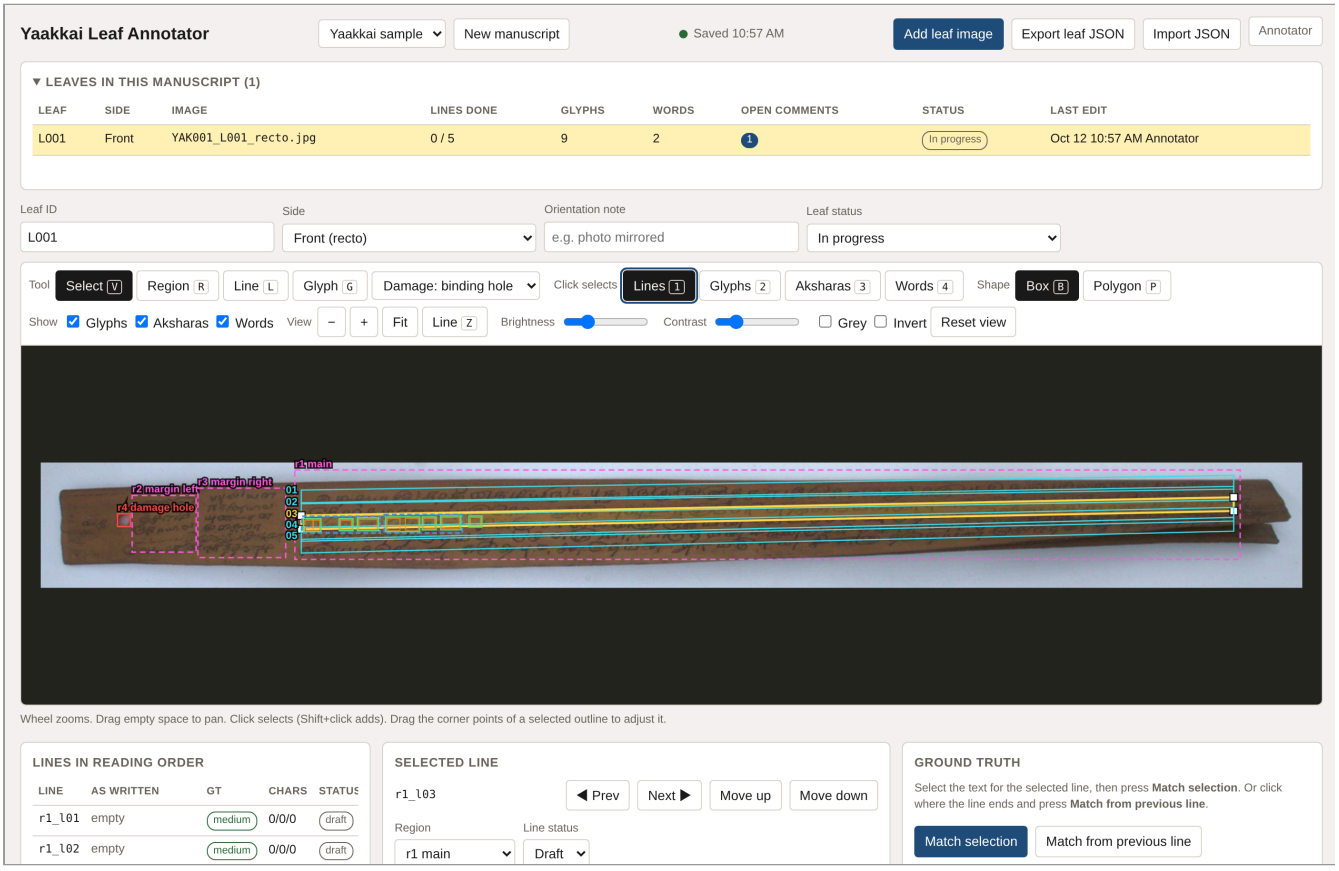


The Account menu

If you forget your password, ask the admin to set a new temporary one.

After you sign in, the tool reopens the manuscript and leaf you worked on last.

3. The screen at a glance



The whole screen with a leaf open

From top to bottom:

1. **Top bar:** the manuscript you are working in, the save indicator, and the main buttons.
2. **Leaves in this manuscript:** every leaf, with its progress. Click the heading to fold it away.
3. **Leaf details:** Leaf ID, Side, Orientation note, Leaf status.
4. **Toolbar:** drawing tools and view controls.
5. **The image:** the leaf with everything drawn on it.
6. **Three panels:** lines and regions on the left, the selected line in the middle, ground truth and checks on the right.
7. **Comments to the developer** and **Edit history** at the bottom.

The top bar

Item	What it does
Manuscript dropdown	Switches between manuscripts

Item	What it does
New manuscript	Creates a manuscript (section 4)
Save indicator	Green “Saved”: all your work is stored. Amber: saving in a moment. Red: read the message under the top bar.
Add leaf image	Uploads a photograph as a new leaf (section 5)
Export leaf JSON	Downloads the open leaf as a file, if the admin asks for one
Import JSON	Only use when the admin asks you to. It replaces everything on the open leaf.
Your name	Opens the Account menu (section 2)

Messages

Messages appear as bars under the top bar. Amber bars give information; red bars need you to act. Some have a button, such as **OK** or **Restore them**. If you ever see “Demo mode”, stop and tell the admin: your work would not be saved.

4. Manuscripts and ground truth

A manuscript is a group of leaves that share one ground truth text (the existing transcription). Choose the manuscript in the dropdown at the top before you do anything else.

Create a manuscript

1. Press **New manuscript**.
2. Type its name or year, for example “Puthur temple record, 1857”.
3. Press **Create**.

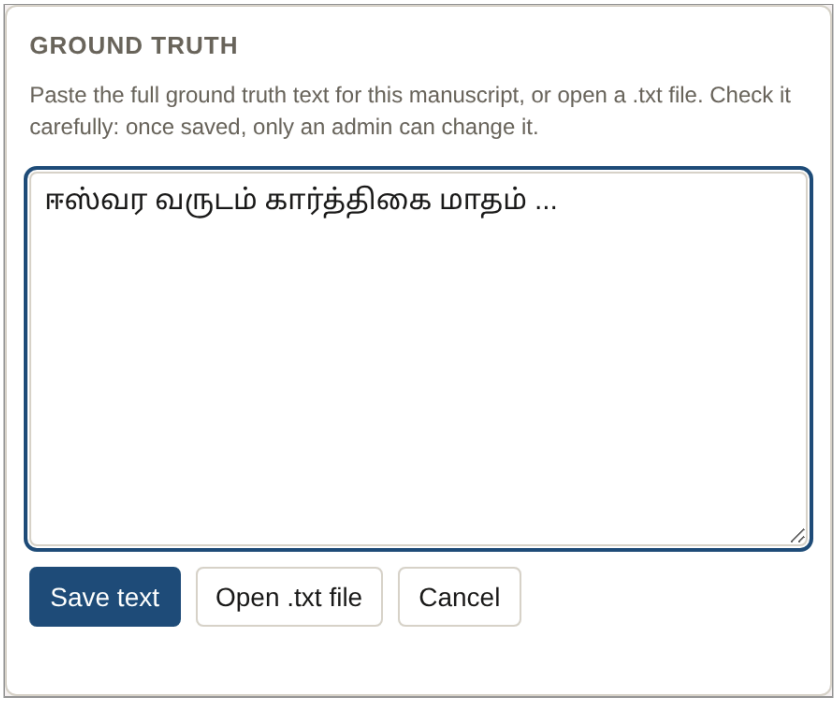


Creating a new manuscript

Add the ground truth

A new manuscript has no ground truth yet, so the **Ground truth** panel shows a text box.

1. Paste the full transcription, or press **Open .txt file**.
2. Read it through once more. **Once saved, only the admin can change it.**
3. Press **Save text**.



Adding ground truth to a new manuscript

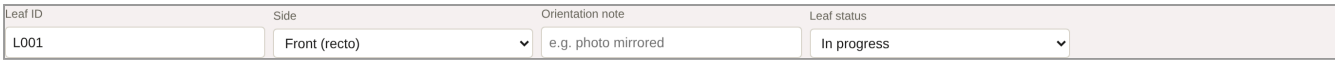
Paste the text exactly as you received it. Do not correct it. If the leaf and the ground truth disagree, record that on the line instead (section 10). If you spot a mistake after saving, leave a comment of type **Ground truth problem** for the admin.

5. Adding and opening leaves

A leaf is one photograph of one side of a palm leaf.

Add a leaf

1. Check the right manuscript is chosen at the top.
2. Press **Add leaf image** and choose the original photograph (JPEG, PNG or WebP).
3. The image uploads and opens.
4. Fill in **Leaf ID** and **Side** straight away.



Leaf details

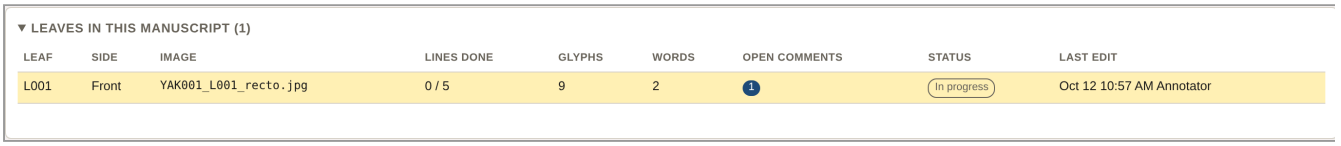
Field	What to enter
Leaf ID	The leaf's catalogue number, for example L003. Front and back of the same leaf share the same Leaf ID.
Side	Front (recto), Back (verso), or Not known
Orientation note	Anything odd about the photo, for example "photo mirrored"
Leaf status	In progress while you work; Ready for review when you finish (section 14)

Good to know:

- Use the file names the admin gives you. If a photo with the same file name is already in the manuscript, the tool opens that leaf instead of adding it again.
- A warning appears if a photo is less than 400 pixels tall. The letters will be too small to read, so ask the admin for the original photograph.
- You cannot swap the photo on an existing leaf. A better photo goes in as a new leaf.

Open a leaf

The **Leaves in this manuscript** list shows every leaf, most recently edited first. Click a row to open it.



The leaves list

The columns show the Leaf ID, side, image file name, lines done out of all lines, glyph and word counts, open comments (in a blue circle), status, and who edited it last.

6. The toolbar and the image



The toolbar

The dark button in each group is the one in use. Letters in small boxes are keyboard shortcuts.

Control	Key	What it does
Select	V	Click things to select them, drag white corner squares to adjust, drag empty space to move the image
Region	R	Draw a region of the type in the dropdown next to it (section 7)
Line	L	Draw around one line of text (section 7)
Glyph	G	Draw a box around one mark (section 9)
Click selects: **Lines / Glyphs / Aksharas / Words**	1 / 2 / 3 / 4	Which kind of thing a click on the image picks
Box / Polygon	B / P	Draw regions and lines as a rectangle, or by clicking points. Finish a polygon with Enter.
Show: Glyphs, Aksharas, Words		Hide or show each kind of box on the image
− / + / Fit	- / + / 0	Zoom out, zoom in, show the whole leaf
Line (in View)	Z	Zoom in on the selected line
Brightness, Contrast, Grey, Invert		Make faint ink easier to see. This never changes the photo itself.

Control	Key	What it does
Reset view		Undo those adjustments

Moving around

- **Mouse wheel:** zoom in or out where the pointer is.
- **Drag empty space** (Select tool): move the image.
- **Hold Space and drag:** move the image while drawing.
- **Shift + click:** add to what is selected.

Colours on the image

Colour	Meaning
Pink, dashed	Text region
Red, shaded	Damage (hole, split)
Cyan	Line, with its number on the left
Orange, dashed	Glyph not yet in an akshara
Green	Akshara, with its text above when zoomed in
Blue, dashed	Word, with its text below when zoomed in
Yellow	Selected

7. Regions and lines

Draw regions first, then the lines inside them.

Regions

1. Choose the **Region** tool (R).
2. Pick the type in the dropdown next to it.
3. Drag a box around the area.

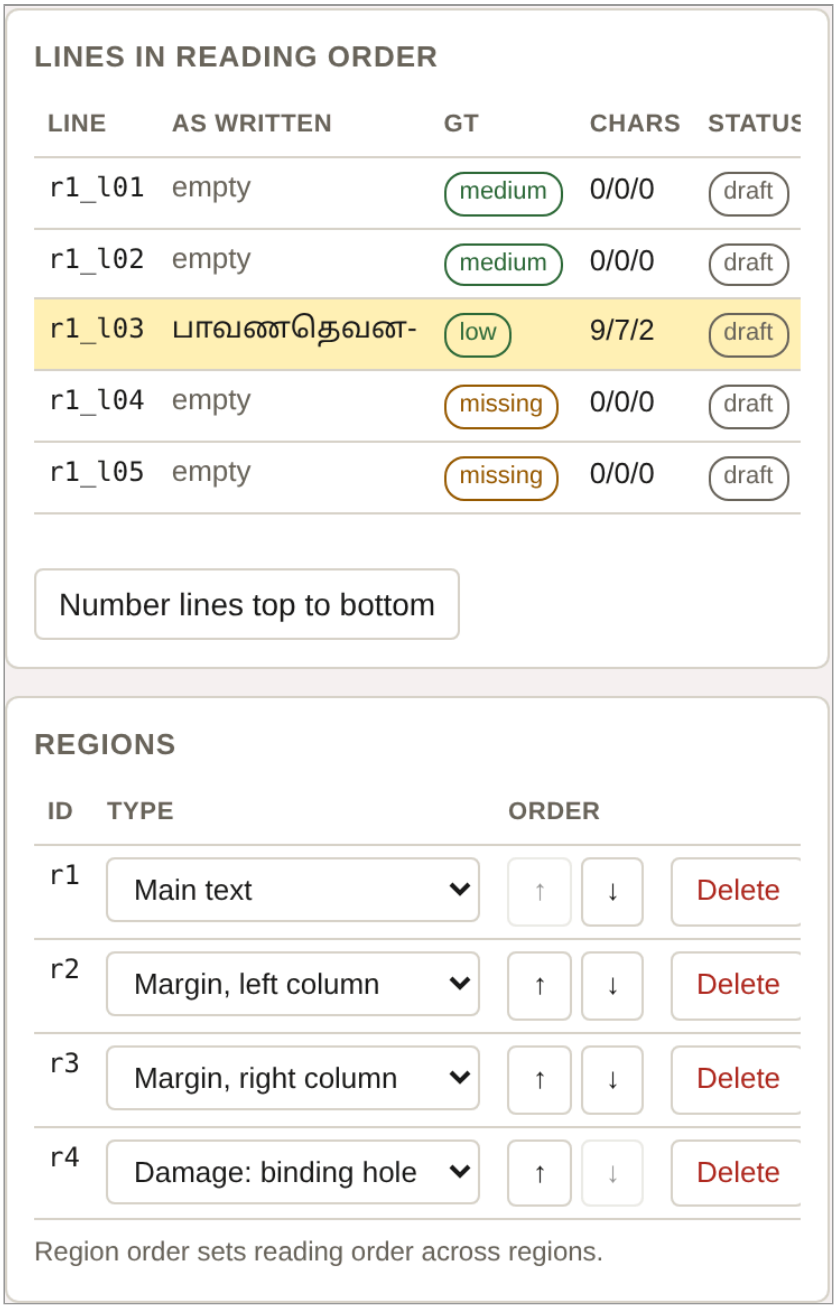
Type	Use for
Main text	The main block of writing

Type	Use for
Margin, left / right / other	Separate blocks in the margins, such as the short columns beside the hole
Symbol or mark	A mark on its own, such as the pillaiyar suzhi
Damage: binding hole / split or break / other	Physical damage. Shown shaded red.

In the **Regions** panel, the ↑ ↓ arrows set the reading order. Lines are read region by region in that order. **Delete** needs a second click to confirm.

Lines

1. Choose the **Line** tool (L).
2. Drag a box around one line of writing. If the line slopes, use **Polygon** (P): click the four corners and press Enter.
3. Repeat for every line.
4. Press **Number lines top to bottom** to put them in order.



The lines and regions panels

Line IDs look like r1_l03: the third line of region r1. Click a line in the list, or on the image, to select it.

The lines list shows:

- **As written:** the start of the line's text.
- **GT:** whether the line is matched to the ground truth, and how sure the match is. Amber "missing" means not matched yet.
- **Chars:** glyphs / aksharas / words on the line.
- **Status:** Draft or Done.

To reshape a line, select it with the **Select** tool and drag its white corner squares. **Move up / Move down** and **Prev / Next** are in the Selected line panel. **Delete line** (at the bottom of that panel) needs a second click, and also deletes the line's glyphs, aksharas and words.

8. Matching lines to the ground truth

Each line is linked to the part of the ground truth text it matches.

1. Select the line.
2. In the **Ground truth** panel, drag across the text for that line.
3. Press **Match selection**.
4. Set **Match confidence** in the Selected line panel: High if you are sure, Medium if fairly sure, Low if it is a guess.

Faster, once the first line is matched: select the next line, click in the ground truth where that line **ends**, and press **Match from previous line**. The match starts where the previous line's match finished.



The ground truth panel with line 3 selected

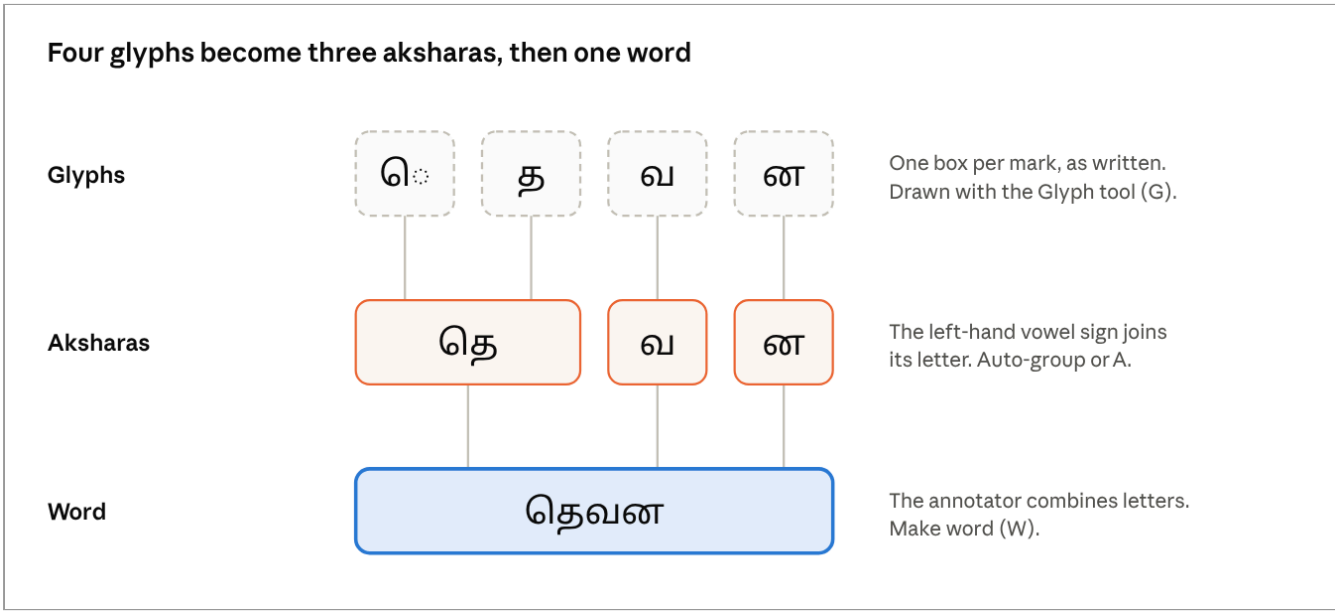
What you see	Meaning
Grey underline	Already matched to a line on this leaf
Small blue number	The line whose match starts there
Yellow	The selected line's match
Red double underline	Two lines claim the same text. Fix one.
Plain text	Not matched on this leaf. Text for the other side of the leaf stays plain.

If a line has no ground truth at all (a number in the margin, a mark), tick **No ground truth for this line** in its flags. **Remove ground truth match** undoes a match.

9. Characters: glyphs, aksharas and words

Each line is broken down into three levels:

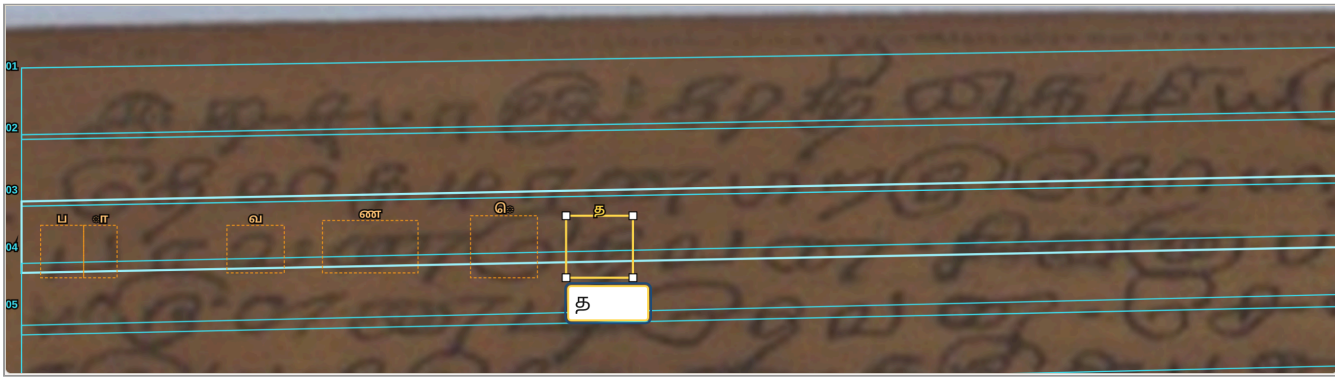
- **Glyph:** one separate mark on the leaf, in its own box.
- **Akshara:** one letter, made from one or more glyphs.
- **Word:** letters you combine into a word.



Four glyphs become three aksharas, then one word

Step 1: draw the glyphs

1. Select the line and press **Z** to zoom in on it.
2. Choose the **Glyph** tool (G).
3. Drag a tight box around one mark. It must sit inside the line's outline.
4. A small typing box appears under it. Type the mark and press **Enter**.
5. Go on to the next mark, left to right.



Drawing a glyph and typing its mark

Rules for glyphs:

- **One mark per box.** A vowel sign written on the left (ெ ே ை) gets its own box, separate from its letter.
- **Joined shapes are one glyph.** Where letter and vowel sign are one shape, as in கு, கூ or டி, draw one box and type the whole letter.
- **Pulli only if you can see it.** Palm leaf scribes usually left it out.
- **Separators are glyphs.** Draw the dashes and dots between names; type - or .
- **Numerals as written.** Use the Tamil numeral keys (க, ய, ஙூ), never 1, 10 or the word "day".
- **Cannot read a mark?** Still draw the box. Leave it empty or type [...], and tick **Illegible**.

To retype a mark later: press **2** (click selects glyphs), click the glyph, then double-click it.

Step 2: group glyphs into aksharas

Press **Auto-group glyphs**. The tool joins each letter with its vowel signs:

- ெ ே ை join the letter on their right.
- ா ி ீ ு ூ ் join the letter on their left.
- Split vowels are combined: ெ + க + ா becomes கொ.

Check every akshara afterwards. The tool cannot tell ா from ர for you. Anything it was unsure of shows in red.

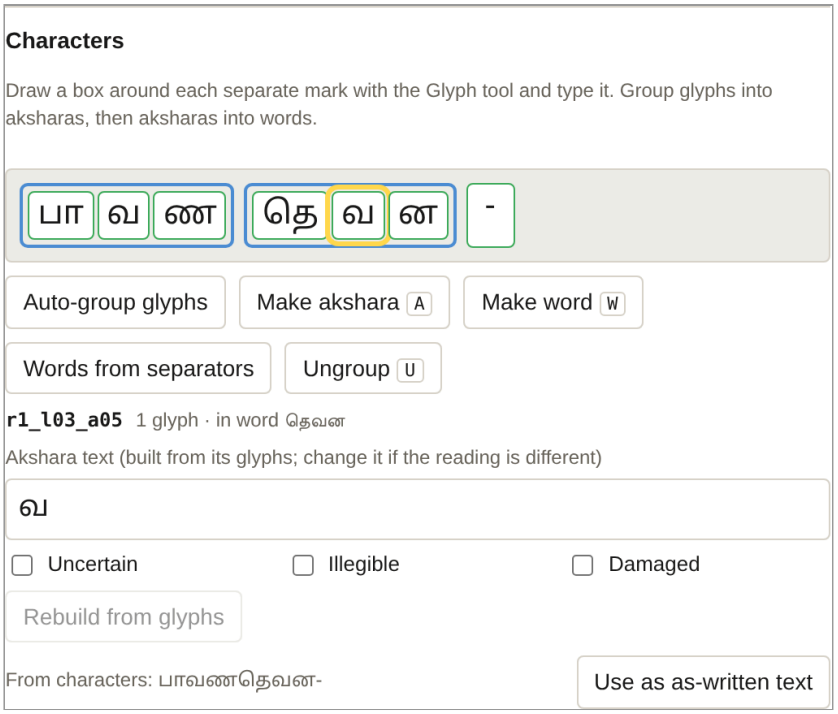
To group by hand: press **2**, click the first glyph, Shift + click the others, then press **Make akshara** (A).

If a letter's reading should be different from its glyphs, type it in **Akshara text**.

Step 3: combine aksharas into words

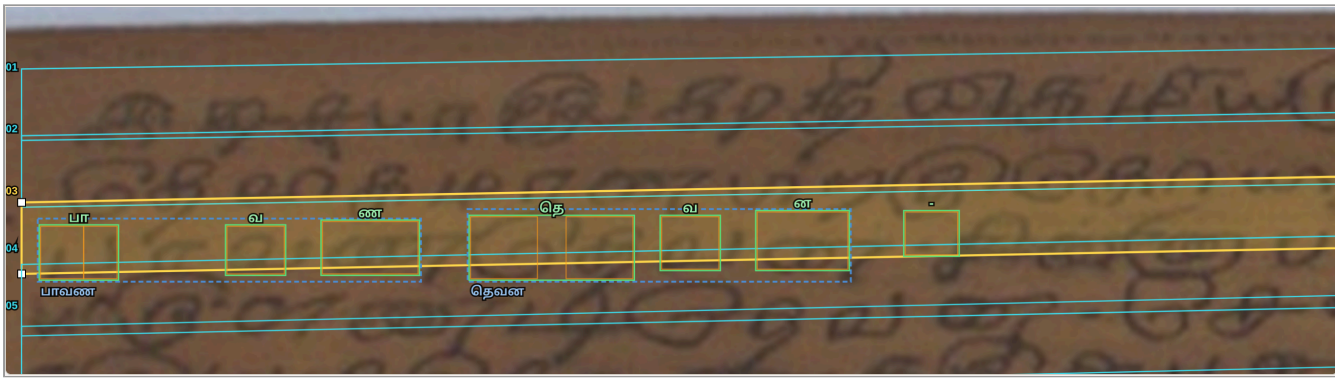
1. Click the first letter of the word in the strip (or press **3** and click it on the image).
2. Shift + click the other letters of the word.
3. Press **Make word** (W).

Words from separators makes words automatically between the dashes and dots. Check each one, then untick **Proposed, not checked** on it.



The Characters section: two words, a separator, and one akshara selected

In the strip, a blue frame is a word, a green box is an akshara, a dashed orange box is a glyph not yet in an akshara, and the yellow outline is what you have selected.



Glyphs, aksharas and words on the image

Undo and delete

- **Ungroup** (U) splits a selected akshara back into glyphs, or a selected word back into letters.
- **Delete** key or **Delete glyph** removes the selected glyph.

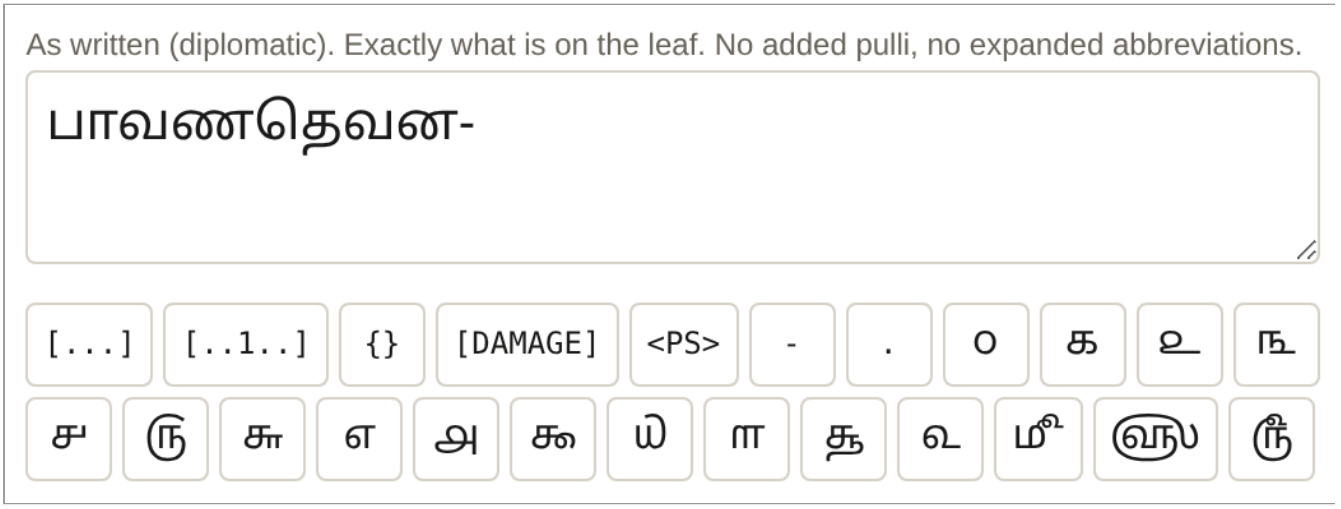
Glyphs and aksharas can be flagged **Uncertain**, **Illegible** or **Damaged**. Words can be flagged **Uncertain**.

10. As-written text, keys, flags and disagreements

As-written text

The **As written** box holds exactly what is on the leaf. Do not add pulli, do not expand abbreviations, and keep the vowel sign the scribe wrote even if it looks wrong.

If you drew the characters, press **Use as as-written text** in the Characters section to fill this box from them.



The as-written box and the marker keys

The keys

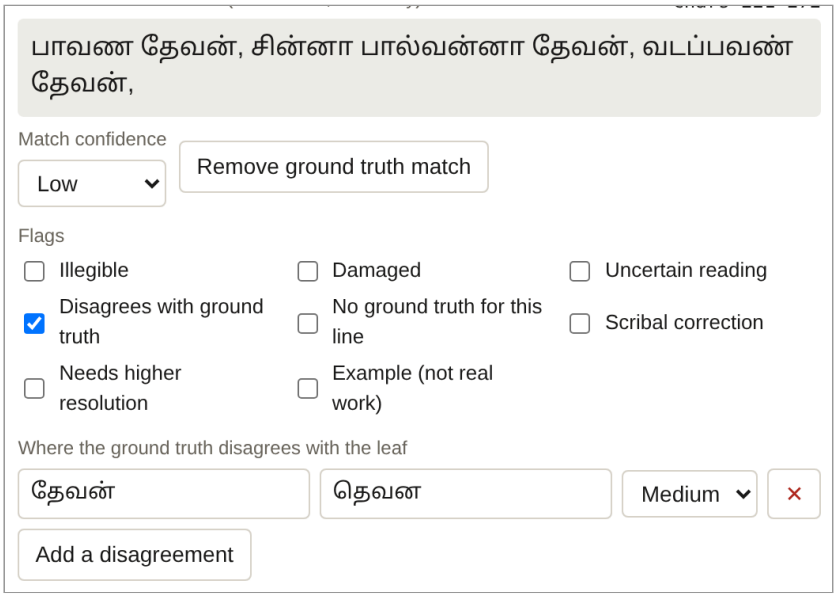
The keys type into the last box you clicked in: the as-written box, a glyph's typing box, akshara or word text, or a comment.

Key	Meaning
[...]	Cannot read; number of letters unknown
[..1..]	Cannot read one letter. Change the number as needed.
{}	Unsure of a letter. Select the letter first and it is wrapped: {ர}
[DAMAGE]	A gap caused by damage
<PS>	Pillaiyar suzhi
- and .	Separators between names
o to கூ	Tamil digits 0 to 9

Key	Meaning
௰ ௱ ௲	Tamil 10, 100, 1000
௳ ௴ ௵ ௹	Day, month, year and rupee signs

Tamil letters and signs opens the full set of letters, for computers without a Tamil keyboard.

Flags and disagreements



Ground truth for the line, flags and a disagreement

Flag	Tick when
Illegible	You cannot read the line at all
Damaged	Damage affects the line
Uncertain reading	Parts of your reading are doubtful
Disagrees with ground truth	The leaf and the ground truth differ. Ticked for you when you add a disagreement.
No ground truth for this line	Nothing in the ground truth matches this line
Scribal correction	The scribe corrected or wrote over something
Needs higher resolution	The photo is too poor to read this line
Example (not real work)	Practice only

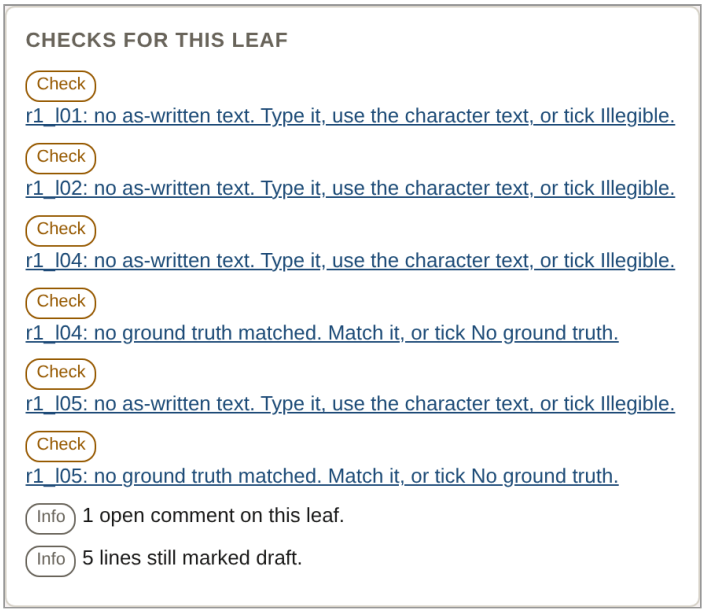
When the leaf and the ground truth differ, press **Add a disagreement**. Type what the ground truth says, what the leaf seems to show, and how sure you are. Add one for each difference. Never change the ground truth itself.

Notes is for anything else about the line. For a question that needs an answer, use a comment instead (section 12).

When the line is finished, set **Line status** to **Done**.

11. The checks panel

Checks for this leaf lists everything still unfinished on the leaf. Click a check to jump to the line or character it is about.



The checks panel

- **Fix** (red): must be fixed before the leaf goes for review.
- **Check** (amber): fix it, or leave a comment explaining why it is right.
- **Info** (grey): for your information.

Check	What to do
Leaf ID or Side empty	Fill in the leaf details
Line has no as-written text	Type it, use the character text, or tick Illegible
Line has no ground truth matched	Match it, or tick No ground truth
Glyphs with no mark	Type the mark, or tick Illegible

Check	What to do
Glyphs not in an akshara	Auto-group, or Make akshara
Glyph box outside the line	Move the box, or the line outline
As-written text differs from the characters	Correct one of them
Two lines use the same ground truth	Re-match one of them
Ground truth edited after matching	Recheck every match on the leaf

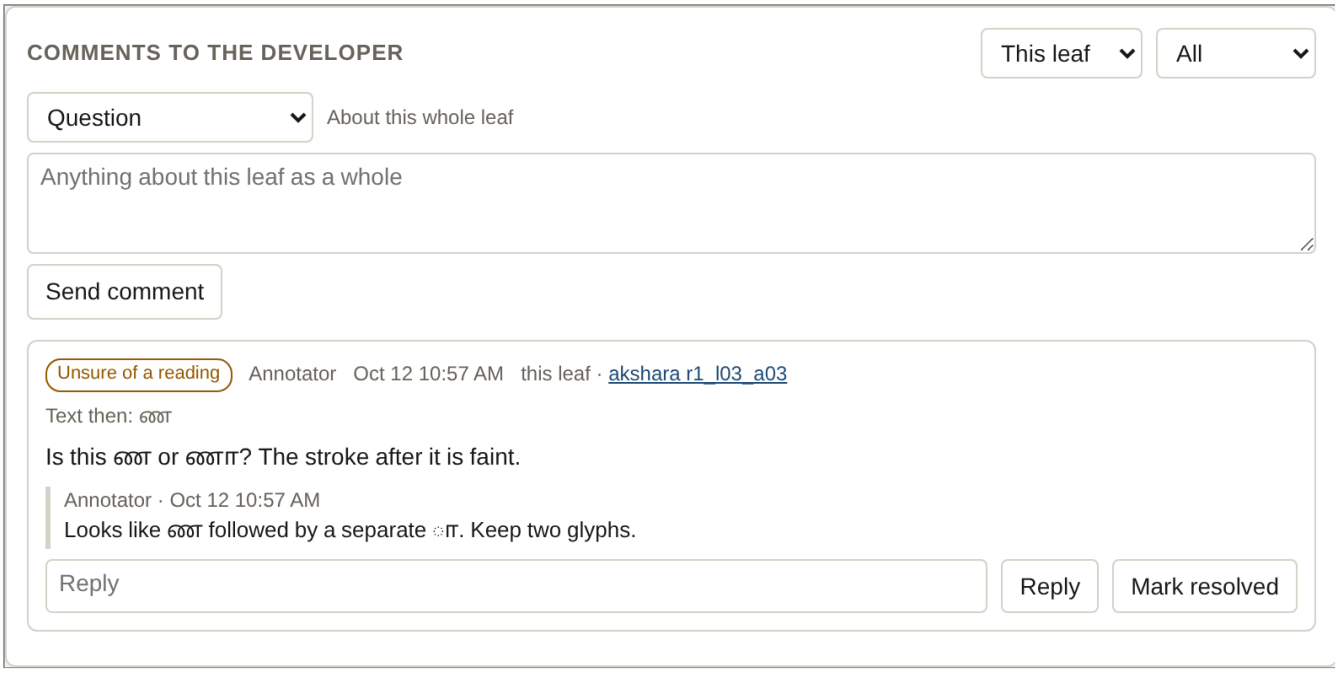
12. Comments to the developer

When you are unsure, do not guess: leave a comment. The developer answers in the same place.

- **About a line, glyph, akshara or word:** select it, then write in **Comment to the developer about ...** in the Selected line panel. The comment is attached to whatever is selected.
- **About the whole leaf:** use the box in **Comments to the developer** at the bottom of the page.

Choose the type, write the comment and press **Send comment**.

Type	Use for
Unsure of a reading	You are not sure what a letter or word is
Ground truth problem	The ground truth looks wrong or has text missing
Image problem	Blurred, cut off, too dark, wrong leaf
Question	How to annotate something
Tool problem	Something in the tool is not working



A comment about one akshara, with a reply

Each comment shows who wrote it, when, and what the text said at that moment. Reply with **Reply**. Press **Mark resolved** when the question is settled. You cannot edit a comment after sending it, so read it before you press Send.

The filters at the top switch between **This leaf** and **All leaves**, and between **Open**, **Resolved** and **All**. A blue number in the lines list or leaves list means there are open comments there.

13. Saving, edit history and conflicts

There is no Save button. Your work saves about a second after each change. Watch the dot at the top:

- **Green, "Saved"**: everything is stored.
- **Amber**: saving in a moment.
- **Red**: not saved yet. The tool keeps trying, and keeps a copy in your browser meanwhile. Do not close the tab while it is red.

If the internet drops and a save does not get through, the next time you open that leaf a message offers **Restore them** (put your unsaved changes back) or **Discard**.

Edit history

Edit history for this leaf, at the bottom of the page, lists every change: when, who, which line, and what changed, with the old value crossed out beside the new one. Nobody can change or delete the history.



The edit history

If someone else edits the same leaf

If another person saves the leaf you are working on:

- **If you have no unsaved changes**, your screen updates with their work by itself.
- **If you do**, saving pauses and a red message names who saved. Choose **Load their version** (your unsaved changes are dropped) or **Keep mine and save over theirs** (their newest changes are replaced).

To avoid this, agree with your colleagues who works on which leaf.

14. Finishing a leaf

1. Every line is marked **Done**.
2. The checks panel has no red **Fix** items. Every amber **Check** is fixed or explained in a comment.
3. Anything you were unsure of has a comment.
4. Set **Leaf status** to **Ready for review**.

A reviewer then checks the leaf and marks it **Reviewed**. Only reviewers and the admin can do that. If you change a leaf after it was reviewed, it goes back to **Ready for review** by itself, so the reviewer sees your change.

If the admin asks you for a copy of a leaf, open it and press **Export leaf JSON**.

15. Keyboard shortcuts

Shortcuts work when you are not typing in a box. If a key does nothing, click on the image first.

Key	Action
V	Select tool
R	Region tool
L	Line tool
G	Glyph tool
B / P	Draw boxes / polygons
1 / 2 / 3 / 4	Clicks select lines / glyphs / aksharas / words
A	Make akshara from selected glyphs
W	Make word from selected aksharas
U	Ungroup selected akshara or word
Delete	Delete selected glyphs
Z	Zoom to the selected line
0	Show the whole leaf
+ and -	Zoom in and out
[and]	Previous and next line
Space + drag	Move the image
Shift + click	Add to selection
Double-click a glyph	Retype its mark
Enter	Finish a polygon, or close the typing box
Esc	Cancel drawing

16. Troubleshooting

What you see	What to do
"Email or password is wrong"	Check the email for typing mistakes. If it still fails, ask the admin to check your account.
"No access" after signing in	Your account has been switched off. Ask the admin.
"Demo mode" message	Stop work and tell the admin. Nothing is being saved.
A photo will not upload	Use JPEG, PNG or WebP. Ask the admin to convert TIFF files.
"The image could not be loaded"	Reload the page.
"Draw glyphs inside a line outline"	Draw the line first, or make its outline bigger.
"Click in a text box first"	Click in the box you want to type into, then use the keys.
Reviewed cannot be chosen	Only reviewers and the admin can choose it. Choose Ready for review.
"Ground truth could not be saved"	This manuscript already has ground truth. Only the admin can change it. Leave a Ground truth problem comment.
"Saving paused"	Someone else saved this leaf. Choose a version (section 13).
Red "Not saved"	Your internet may have dropped. Keep the tab open; the tool keeps trying.
Tamil letters show as empty boxes	Check your internet and reload the page.

For anything else, leave a **Tool problem** comment saying what you did and what you saw.